

Agentic trading system Engineering report

How the build is structured: control plane, signal pipeline, fail-closed gates, and the split between supervised chat and one unsupervised Automation.

Owner: Jarrod Besner · Broker: Robinhood · Tradable account: Agentic •••2907 · Repo: Github-agentic-repository- · As of 2026-08-30

Options playbook RELEASED

Agent H Automation ON

This chat still confirm-gated

0 live places from this build

13 unit tests passing

What this system had to solve

Product constraint

- One funded account, \$1,500 buying power
- Options first, equity fallback
- Max one open position, one contract
- Watchlists as the only universe
- No crypto, no shorts, no multi-leg

Engineering constraint

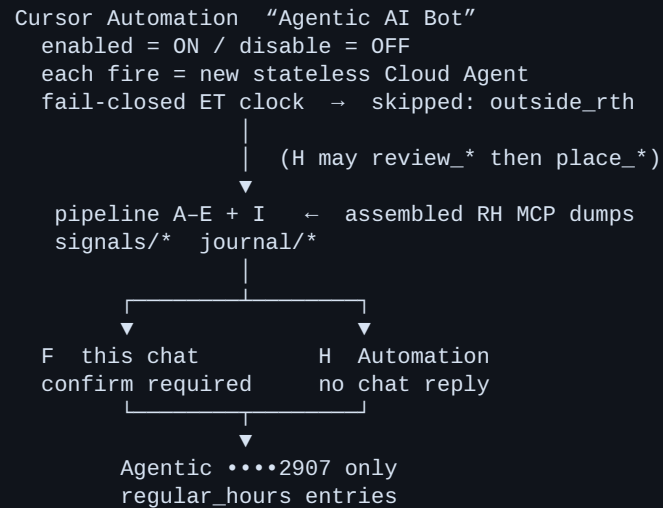
- Broker access is Robinhood MCP, not a custom OMS
- No OCO / bracket orders in the tool set
- Cursor Automations are *stateless discrete fires*
- Cloud agents can invent numbers unless gated
- MCP exposes every tool once connected

The build is a gated decision pipeline plus two place authorities — not a continuously running trading server.

Fail-closed, honest numbers, one kill switch

1. **Fail closed.** Missing clock, account, quote, delta, spread, or buying power → place nothing.
2. **Do not invent market data.** Greeks only from `get_option_quotes`. No Black-Scholes fallback.
3. **Spec is code.** Locked rules live in `config/rules.json` and `config/autonomous_permissions.json`.
4. **Agents are modules, not fleets.** A-E + I are functions in one pipeline. One Automation (H) may place.
5. **Two authorities, never mixed.** This chat needs a specific-order confirm. H has standing permission only while the Automation is enabled.
6. **Mask secrets in git.** Full account numbers only in RH tool args. Journal and reports use `••••2907`.

Two place paths, one account



Connecting Robinhood MCP grants every broker tool. Forbidden calls are prompt-enforced, not tool-stripped. Disable is the kill switch.

Letters are pipeline stages

ID	Role	Code	Runtime
A	Watchlists, drop crypto, ADV $\geq$ 2M	pipeline/universe.py	Function in cycle
B	Pattern heuristics + majority bias	pipeline/patterns.py	Function in cycle
C	News / earnings packaging	pipeline/news.py	Function in cycle
D	Long call/put, DTE 0–7, ATM – 1 OTM	pipeline/options_structure.py	Function in cycle
I	Copy RH Greeks; $ \Delta $ 0.40–0.50	pipeline/greeks.py	Function in cycle
E	–20% / +40% cash-risk math	pipeline/risk.py	Function in cycle
F	Supervised review / place gate	pipeline/execution.py	This chat
G	One-shot orchestrator	pipeline/orchestrator.py	Script, not a daemon
H	Unsupervised RTH cycle + place	playbooks/agent_h_autonomous.PROMPT.md	Cursor Automation

Do not spin A–D into separate Automations. They share one universe and one bias. Two place-capable agents on one account can double-fill.

MCP in, deterministic filters out

```

RH MCP (read)
  watchlists · fundamentals · historicals
  chains · instruments (paged) · quotes · portfolio
    |
    ▼
  data/raw/*.json    assembled by the Cursor agent
    |
    ▼
  scripts/run_phase2_cycle.py
  pipeline.orchestrator.run_pipeline(raw)
    |
    ▼
  signals/universe.json
  signals/technicals.json
  signals/option_candidates.json
  signals/greeks.json
  signals/risk_plan.json
  journal/YYYY-MM-DD.md
    |
    ▼
  F confirm or H review_* → place_*
  (Phase 2/3 code paths never place)

```

`run_pipeline` asserts `phase2_places_orders` is `False`. Place is a later human or H step, not a pipeline side effect.

Locks are files, not chat memory

config/rules.json

- Account last-4, universe, crypto off
- Pattern types and timeframes
- DTE, delta band, 1 contract
- Spread prefer 5% / reject 10%
- Working pair -20% / +40%
- Playbook RELEASED
- F still confirm-gated
- unsupervised_agent_h: enabled

config/autonomous_permissions.json

- Status ACTIVE as of 2026-08-30
- Allowed: review/place/cancel equity + options
- Forbidden: crypto, exercise
- RTH-only scan and buy
- Max one open position
- Automation id recorded after first fire
- Missing file → H must place nothing

Later owner locks supersede earlier ones. Journal keeps the history; the JSON is the current machine spec.

All lists, then drop, then ADV

A

extract_watchlist_symbols

- Union every Robinhood list
- Include Options Watchlist underlyings
- Drop currency_pair / tokenized stock
- Dedupe by symbol

2M

min average_volume

- From get_equity_fundamentals
- Batches of ≤ 10 symbols
- Missing volume $\rightarrow$ reject
- Weekend scan: 89 $\rightarrow$ 72 liquid

0

crypto path

- BTC / ETH / XRP skipped
- place_crypto_* forbidden
- Index underlyings allowed
- No second universe file

Deterministic heuristics, strict majority

detect_patterns

- Needs ≥ 30 OHLC bars
- Local extrema, order = 3
- H&S, inverse H&S, double/triple top & bottom, triangles
- ~1.5% near-equality on peaks/troughs
- Each hit carries bullish or bearish

dominant_bias

- Count bullish vs bearish hits
- Require **strict majority**
- Tie or empty $\rightarrow$ null $\rightarrow$ no option structure
- H cost control: daily first; 10m + 1h only on daily hits; stop at one bias

RH MCP has no 15minute interval. Closest bar is 10minute. Rules JSON still lists 15m as the intended TF; live scans used 10m.

Bias → one vanilla long

Step	Rule	Implementation
Side	Bullish → long call. Bearish → long put. Else skip.	<code>choose_structure_from_bias</code>
Expiry	Calendar DTE 0–7. Do not guess DTE.	<code>filter_expirations</code>
Strike	ATM if within 1% of spot, else one OTM.	<code>pick_atm_or_one_otm</code>
Size	Exactly 1 contract. Keyword-only risk math.	<code>options_risk_plan(..., contracts=1)</code>
Hours	<code>market_hours=regular_hours</code> on entries.	F proposal + H prompt

Instrument lists paginate at 100. First-page “OTM” on AMD/MU was a pagination artifact. H and F must page until ATM is in set.

Greeks and spread refuse invention

extract_greeks

- Copies delta, gamma, theta, vega, rho, IV only if present
- Missing fields listed, never filled
- `delta_in_band` uses `abs(delta)` 0.40–0.50
- 0.396 (NU) and 0.522 (META) both fail — the band is closed

option_quote_liquid

- $\text{mid} = (\text{bid} + \text{ask}) / 2$
- $\text{spread\%} = (\text{ask} - \text{bid}) / \text{mid}$
- Prefer $\leq 5\%$, accept $\leq 10\%$, reject $> 10\%$
- Reject one-sided or $\text{mid} \leq 0$
- **No \$0.10 absolute override**

Cheap names fail on percent width even when the dollar spread looks small. That is intentional.

Math in-repo; broker has no OCO

options_risk_plan

- Keyword-only premium_per_share
- Cash = premium $\times$ 100 $\times$ 1
- Locked pair: SL 20%, TP 40% $\rightarrow$ 1:2
- Bands enforced: SL 20–50%, TP $\geq$ 30%
- Raises if contracts $\neq$ 1

Stop-first until OCO exists

- MCP cannot rest a true OCO
- Two live exits can double-fill
- After fill: broker STOP only
- TP watched on a later RTH fire
- Equity fallback: +25% / –20% of cost

SNOW Sep 4 327.5 put cleared spread and delta on Friday marks, then failed affordability: \$1,975 cash vs \$1,500 buying power.

F is gated in code. H is gated by prompt + enable.

	Agent F (this chat)	Agent H (Automation)
Permission	can_place_live(confirm, released)	Standing prompt + enabled Automation + permissions file
Confirm	Explicit specific order in this chat	No chat reply
Clock	Owner-requested tests may scan Sunday	ET RTH check <i>before any RH call</i>
Place tools	Blocked unless both gates true	review_* then place_* on2907
Off switch	Don't confirm	Disable Automation, or delete permissions file

Proposal builder hard-codes quantity 1, GFD, regular_hours, buy-to-open, and places_order: false.

Automation treats cron as untrusted

Clock first

- America/New_York
- Mon–Fri 09:30 inclusive → 16:00 exclusive
- Weekend / holiday / pre / after → journal one line, exit
- Do not invent a holiday calendar
- If clock says RTH but RH session is closed → still skip

Proved on first fire

- Automation Agentic AI Bot: enabled
- Run “Automated trading logic”
- Sunday 06:49:57 EDT
- Wrote skipped: `outside_rth`
- No RH calls, no scan, no buy

Cron can still start a billable run at night. The prompt exits; it does not delay until the open.

The MCP is the integration surface

Limit	Engineering response
No 15minute historical interval	Use 10minute + hour + day
Instruments page size 100	Follow next cursor; never treat first-page max strike as ATM
Fundamentals batch ≤ 10	Chunk watchlist symbols
Quotes >20 omit official closes	Keep quote batches small when close is required
No OCO	Stop-first; TP on a later run
Buying power is not on accounts	<code>get_portfolio</code> is authoritative spendable cash
All tools exposed with the server	Prompt denylist + Disable kill switch

Small surface, explicit artifacts

Runtime

- `pipeline/*.py` — A-G + I
- `scripts/run_phase2_cycle.py`
- `pipeline/tests/` — 13 tests
- `data/raw/` — assembled MCP dumps
- `signals/` — last cycle JSON
- `journal/` — human locks + skips

Spec and docs

- `config/rules.json`
- `config/autonomous_permissions.json`
- `playbooks/*.md` + H prompt
- `docs/phase0-trading-plan-printable.html`
- `docs/system-creation-report-printable.html`
- `docs/engineering-presentation.html` + `.pdf`

Phase 4 “always-on loop” is not a process in the repo. Discrete Automation fires replace a daemon.

Unit tests lock the gates

Test	What it protects
Crypto exclusion	BTC-USD never enters equity universe
ADV gate	Below 2M rejected
Spread 5 / 10 / reject	Preferred, acceptable, too wide; cheap \$0.10 still rejects
Greeks no-invention	Missing theta stays missing
Risk math + bands	-20/+40 on \$2 = \$200 / \$160 / \$280; illegal SL/TP raise
ATM + DTE filter	As-of Sunday, Sep 2 is in window
F place gate	Draft or no confirm → cannot place

Last recorded: `python3 -m pytest pipeline/tests -q` → 13 passed. Live RH cycles are journaled separately; they are not mocked in CI.

The weekend scan exercised the whole funnel

89 → 72

equities after ADV

17 names failed 2M volume. Crypto already dropped.

28

combined-bias names

16 call / 12 put. CPRT and VIRT had no 0-7 DTE weekly.

3

stale-quote passers

GOOGL call, TSM put, SKHY put. SNOW failed cash. 0 orders.

Engineering takeaway: the bottleneck is gates (spread, delta, DTE, cash), not agent count. Pagination and the missing 15m interval are the adapter bugs that matter.

What is unfinished on purpose

Not built

- Always-on Phase 4 daemon
- Broker OCO / brackets
- Holiday calendar service
- Tool-level MCP denylist (prompt only)
- Bar-key adapter: RH uses `*_price`; detector wants `close`

Operational risk

- H first fire branched from `main` and opened a one-line journal PR — do not merge over the full journal
- Two H instances can race if duplicated
- Weekend/night cron still bills a skip run
- signals/* snapshots can lag current locks

Built as a gated pipeline. Live place is still rare.

On

- Spec files + playbooks
- Deterministic A-E + I pipeline
- F confirm gate in code
- H Automation enabled; Sunday skip proved
- Printable creation report + this deck

Off / unused

- No live place_* from this build
- This chat still needs a named ticket
- First legal H entry window: Monday RTH
- Companion Letter report: docs/system-creation-report-printable.html

Engineering summary: one account, one contract, honest quotes, fail-closed session, stop-first exits, and a single unsupervised Automation — not a fleet.